PRASHANT YADAV

Chapter Lead — AI Development & Delivery | BHP India Technical Centre via TCS | AI Architecture | Platform Engineering | DevSecOps | Agentic AI | Multi-Cloud

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PROFESSIONAL SUMMARY

Senior technology leader with 17 years of experience spanning cloud infrastructure, platform engineering, DevSecOps, AI architecture, and delivery management. Currently Chapter Lead AI Development and Acting Delivery Lead at BHP India Technical Centre (BTC), deployed via TCS — leading the AI engineering chapter from the ground up while owning delivery continuity for BTC's key operating model initiatives. Expertise in multi-cloud AI architecture (AWS and Azure, actively building GCP), agentic AI and RAG systems, Kubernetes platform engineering, DevSecOps, and stakeholder management across global enterprise environments. International conference speaker. ISO 42001 certified. MBA in Business Analytics.

KEY EXPERTISE

AI Architecture and GenAI: RAG Pipelines, Agentic AI, Model Context Protocol (MCP), LLMs, Multi-Agent Orchestration, Vector Search, Vertex AI, AWS Bedrock, Azure OpenAI, Gemini, Prompt Engineering, ISO 42001 AIMS, LangGraph, LangChain

Platform and Cloud Engineering: Kubernetes (EKS, Nutanix HCI, Multi-tenant), Docker, Helm, Terraform, IaC, CI/CD, AWS, Azure, GCP, Jenkins, GitHub Actions, Azure DevOps, SonarQube, Trivy, Nexus, Platform Engineering

Leadership and Delivery: Chapter Leadership, Delivery Management, Stakeholder Engagement, People Development, Architecture Governance, RFP Leadership, Vendor Evaluation, Executive Communication

DevSecOps: Jenkins, GitHub Actions, SonarQube, Trivy, Nexus, Air-gapped DevSecOps, RBAC, LDAP, VAPT, ARM Cross-compilation, Security Governance, DRDO Defense Infrastructure

Data Architecture: Medallion Lakehouse, Azure Data Factory, ADLS Gen2, Synapse, Microsoft Fabric, AWS Glue, Redshift, Kafka, Kinesis, MQTT, CDC, Schema Evolution, Data Lineage, Purview, MDM

Security and Compliance: IAM, RBAC, KMS, Key Vault, Zero Trust, PII Masking, Security Trimming, VPC Isolation, Private Endpoints, Audit Logging, ISO 42001, VAPT, Air-gapped Environments

Observability and SRE: Prometheus, Grafana, CloudWatch, Azure Monitor, Log Analytics, ELK/OpenSearch, Application Insights, SLO/SLA Design, Disaster Recovery, Incident Readiness

Edge AI and IoT: TensorFlow Lite, ARM Cortex, AWS IoT Core, Greengrass, MQTT, EfficientDet-Lite, Real-time Edge Inference, Embedded Systems Integration

FinOps: Cloud Cost Optimisation, Resource Rightsizing, Caching Strategy, Cost Attribution per Tenant, Quota Management

Programming and BI: Python (AI Pipelines, Automation, Scripting), React.js, Power BI

WORK EXPERIENCE

BHP India Technical Centre (via TCS / Zone IT Solutions) | Gurgaon, India

Chapter Lead — AI Development & Acting Delivery Lead *Aug 2026 – Present*

BHP's India Technical Centre (BTC) is a global capability centre providing AI and technology development resources to BHP teams worldwide, with work scoped and delivered against business cost centres through a structured operating model.

- **Chapter Lead:** Building BHP India's AI engineering chapter from the ground up — hiring, mentoring, and developing AI engineers; setting engineering standards, coding practices, review governance, and technical capability roadmaps for the chapter.
- **AI Architecture:** Driving design, development, and deployment of GenAI, agentic AI, RAG, and ML solutions across BTC squads; defining AI engineering best practices and promoting GCP Vertex AI, AWS Bedrock, and Azure OpenAI adoption across the centre.
- **Delivery Leadership:** Proactively took ownership of BTC India delivery continuity when the India Delivery Manager departed — now serving as single point of contact for delivery accountability across BTC's key programmes.
- **Vespa:** Leading stakeholder engagement for Vespa — the operating model initiative defining how BHP global demand managers raise resource requests, how BTC allocates engineers, and how delivered work is scoped, tracked, and charged to business cost centres.
- **Turmeric Tool Connect:** Managing Turmeric Tool Connect — the AI-powered interface used by demand managers globally to access BTC capabilities; coordinating with global BHP stakeholders to ensure delivery alignment and resource planning.
- **Capability Planning:** Skills forecasting for the AI chapter 12-18 months ahead; converting bench capacity into capability investment through structured learning programmes and POC initiatives.
- **Agentic AI Agents for Internal Tooling:** Designing and developing internal agentic AI agents to automate and enhance BHP's internal tooling workflows — enabling engineering teams, demand managers, and operations staff to interact with enterprise systems through intelligent, task-driven AI interfaces.
- **myBOS Coach Integration:** Leading integration of the multi-agent AI system into myBOS Coach (BHP Operating System) — embedding AI-powered coaching, workflow automation, and natural-language interfaces directly into BHP's enterprise operating platform to drive adoption of the BOS framework across global operations.

Judge India Solutions Pvt. Ltd. (The Judge Group) | Noida, India

Senior Solutions Architect | Employee #002 *Aug 2016 – Aug 2026*

Growth Path: System Administrator (2016–17) → Cloud Support Engineer (2017–18) → Solutions Architect (2018–20) → Senior Solutions Architect (2020–26)

- Architected enterprise data fusion platforms on Azure (ADF, ADLS Gen2, Synapse, Document Intelligence, AI Search, OpenAI) unifying finance, staffing, payroll, timesheet, and document data into governed intelligence layers.
- Designed agentic AI and RAG GenAI solutions using AWS Bedrock, Azure OpenAI, and vector databases; built multi-agent orchestration platforms with ReAct reasoning loops, HITL gates, Step Functions governance, and immutable audit trails.
- Built Model Context Protocol (MCP) tool orchestration hub enabling dynamic capability discovery and multi-agent tool chaining; reduced end-to-end chain latency from 870ms to 380ms through parallel tool execution and session caching.
- Architected multi-tenant Kubernetes platform on Nutanix HCI supporting 1M+ learners across 76 universities for ICAR national LMS — including AI content generation layer (Bedrock, Polly, Transcribe, Translate, Rekognition).
- Led DevSecOps transformation at DRDO — designed air-gapped DevSecOps ecosystem using Tuleap, Jenkins, Gitea, SonarQube, Nexus with RBAC, LDAP governance, ARM cross-compilation pipelines, and audit logging.
- Delivered CARLA-based automotive digital twin platform for Siemens CES — designed four-layer integration (simulation, middleware, embedded, HMI); reduced provisioning time 70% through Terraform IaC and Jenkins CI/CD.
- Designed Safelite L1 health check automation — six-layer system (Playwright UI, REST health, IMAP mailbox, pixel-match visual state, ServiceNow INC creation, SSH recovery) achieving 90-second full detection-to-recovery cycle across 8 systems.
- Built ARM edge AI surveillance platform for Visionext integrating AWS IoT Core, Greengrass, Lambda, and SageMaker; achieved 28ms on-device inference latency using EfficientDet-Lite on ARM hardware.
- Led ISO 42001 AIMS certification — established AI policy, data quality assessments, VAPT, risk registers, HITL governance frameworks, and BCP artefacts for enterprise AI systems.
- Acted as architecture design authority across POCs, RFPs, and enterprise implementations; drove cloud cost optimisation through rightsizing, caching strategies, and cost attribution achieving measurable spend reductions.

Multiple Organizations | Noida, India

Independent IT Consultant — Infrastructure and BPO Setup *Aug 2012 – Aug 2016*

- Delivered complete BPO infrastructure architecture including dialers, VOIP systems, and networking hardware for multiple startups; directed backend technology operations as Working Director overseeing 15-20 member technical teams.

Dell Perot Systems TSI India Pvt. Ltd. | Noida, India

Sr. Client Support Associate *Jun 2010 – Aug 2012*

- Delivered enterprise IT asset and application support for Dell employees; managed BMC Remedy incident management with SLA-driven resolution workflows; transitioned US hardware support operations to India.

Cyfuture India Pvt. Ltd. / iGATE Global Solutions | Noida, India

L2 Technician / Sr. Technical Support Analyst *Feb 2009 – May 2010*

- Supervised L1 resources managing PPC and SEO technical sales; provided enterprise technical support for Royal Bank of Canada internal users.

KEY PROJECTS

Data Bridge Platform — Azure Enterprise Data Fusion

- Architected Azure-based enterprise data fusion platform unifying fragmented finance, staffing, HR, payroll, timesheets, CRM, invoices, purchase orders, and contracts into a single governed intelligence layer built on Azure Data Lake Storage Gen2 with Raw, Processed, and Curated zones.
- Orchestrated multi-source ingestion via Azure Data Factory with parameterized pipelines, watermark and CDC-based incremental loads, and event-driven triggers; integrated Azure Document Intelligence for OCR extraction of invoices, purchase orders, and contracts using pre-built and custom-trained models.
- Implemented ELT transformations in Azure Synapse and Microsoft Fabric covering entity resolution, master data alignment, schema evolution, and curated business datasets — Resource Allocation, Billing Status, Invoice Matching, Project Margin, and Payroll Reconciliation.
- Built natural-language query intelligence using Azure AI Search (hybrid BM25 + vector + semantic ranking) and Azure OpenAI with citation-grounded responses and role-aware access; delivered Power BI dashboards on the curated layer for billing pipelines, resource utilisation, and margin analysis.
- Established enterprise security with Entra ID, Managed Identities, Private Endpoints, RBAC, Key Vault, PII masking, Azure Monitor audit trails, and data governance with Purview metadata cataloging and end-to-end lineage.
- Stack: Azure Data Factory, ADLS Gen2, Azure Document Intelligence, Azure Synapse, Microsoft Fabric, Azure AI Search, Azure OpenAI, Azure Purview, Power BI, Entra ID, Key Vault, Azure Monitor.

HR Policy RAG Assistant — Enterprise Azure GenAI

- Architected enterprise-grade RAG solution on Azure enabling employees to query HR policies, leave rules, benefits, SOPs, and onboarding guidelines with cited, role-aware, hallucination-guarded responses — 0.91 faithfulness score, 85% auto-resolution rate.
- Built two-pipeline architecture: ingestion pipeline (Azure Blob Storage event triggers → Azure Functions → Document Intelligence OCR extraction) and query pipeline (Entra ID auth → query normalisation → embedding → hybrid retrieval → security trimming → grounded LLM generation).

- Implemented structure-aware chunking (300–500 tokens, 50–80 token overlap) respecting section boundaries; enriched chunks with metadata for policy category, section, effective date, department, and access classification; semantic embeddings via Azure OpenAI (text-embedding-ada-002) indexed in Azure AI Search with HNSW vector search.
- Designed version retention for historical policy auditability, faithfulness gate at 0.80, prompt injection mitigation, low-temperature responses (0.1–0.2) for compliance accuracy, and fallback responses for insufficient evidence to prevent hallucination.
- Stack: Azure Blob Storage, Azure Functions, Document Intelligence, Azure OpenAI, Azure AI Search (hybrid + semantic ranking), Entra ID, Application Insights.

LMS AI Content Generation Platform — AWS Agentic AI

- Architected AI enrichment layer for ICAR national LMS (1M+ learners across 76 universities) — AWS Bedrock agentic platform with ReAct reasoning loop, HITL gates, and Step Functions governance for structured multi-step content generation.
- Built multi-lingual content generation pipeline using Amazon Polly (TTS), Transcribe, Translate, and Rekognition across 6 regional Indian languages; served via multi-tenant Kubernetes platform on Nutanix HCI with tenant isolation enforced at platform level.
- Stack: AWS Bedrock, Amazon Polly, Transcribe, Translate, Rekognition, SageMaker, OpenSearch, Textract, EKS/Nutanix HCI.

Enterprise MCP Tool Orchestration Hub

- Built MCP server exposing AI capabilities as dynamically discoverable tools across multi-agent systems; reduced 3-tool chain latency from 870ms to 380ms via parallel tool execution and session caching; tenant isolation enforced per-tool-call via JWT validation.
- Stack: Python, FastAPI, MCP Protocol, AWS Bedrock, DynamoDB, Kubernetes.

GCP Document Intelligence Agent — RAG + Agentic AI (POC)

- Built hands-on GCP RAG and agentic AI system: Cloud Storage ingestion → Document AI OCR extraction (auto-split into ≤28-page batches) → Vertex AI Gemini Embedding pipeline (text-embedding-005, 768-dim) → Vector Search index (STREAM_UPDATE, COSINE_DISTANCE, Tree-AH) → Gemini 2.5 Flash grounded generation with two-layer confidence filtering (deterministic keyword scan + LLM judgment) — 548 chunks indexed from 1,433 pages of scanned documents.
- 3-agent orchestration: Cache Agent (Firestore persistent lookup) → Search Agent (vector retrieval + confidence check) → Fallback Agent (Gemini general knowledge, clearly labelled); Eventarc-triggered Cloud Functions 2nd gen for auto-ingestion; full per-stage latency instrumentation.
- Stack: Cloud Storage, Document AI, Vertex AI Vector Search, Gemini 2.5 Flash, Cloud Run, Cloud Functions 2nd gen, Eventarc, Firestore, Pub/Sub, IAM.

Siemens CES — CARLA Automotive Digital Twin

- Designed four-layer automotive digital twin (simulation, middleware, embedded, HMI); solved hardware-dependent embedded component challenges via CARLA plugin architecture; reduced provisioning time 70% through Terraform IaC and Jenkins CI/CD. Presented at Embedded World Nuremberg 2025.
- Stack: CARLA, AGL, Android Automotive, Qt HMI, AWS EC2, Terraform, Jenkins, Docker, Kubernetes.

DRDO — Air-Gapped DevSecOps Platform

- Designed secure air-gapped DevSecOps ecosystem for defense infrastructure — Tuleap ALM, Jenkins, Gitea, SonarQube, Nexus with LDAP-based RBAC, ARM cross-compilation pipelines, audit logging, and SCM governance for classified software delivery.
- Stack: Tuleap, Jenkins, Gitea, SonarQube, Nexus, ARM GNU toolchain, LDAP.

Visionext — ARM Edge AI Surveillance Platform

- Designed edge AI pipeline achieving 28ms on-device inference latency on ARM Cortex using EfficientDet-Lite; MQTT event streaming to AWS IoT Core with SageMaker real-time analytics and Greengrass fleet management for distributed camera node updates.
- Stack: ARM Cortex, TensorFlow Lite, EfficientDet-Lite, AWS IoT Core, IoT Greengrass, Lambda, SageMaker, MQTT.

Safelite — L1 Health Check Automation

- Six-layer automated monitoring and recovery system (Playwright UI, Axios REST health, IMAP mailbox check, pixel-match visual state, ServiceNow INC auto-creation, SSH recovery) replacing manual L1 checks across 8 critical systems — 90-second full detection-to-recovery cycle.

FinOps Cloud Cost Investigator — Multi-Agent LangGraph (POC)

- Built a LangGraph multi-agent POC in which a supervisor routes between a tool-calling analyst, a deterministic anomaly detector and a recommender to investigate cloud cost spikes and idle resources; every figure is computed in pandas, never by the LLM.
- Human-in-the-loop approval with interrupt() and checkpointed state: remediation scripts are drafted only after approval, allow-listed and never executed. Guardrails in code (validated routing, loop caps, fallbacks); 13 offline tests with a scripted fake LLM.
- Stack: LangGraph, LangChain Core, Gemini 2.5 Flash, Python, pandas, Pydantic, pytest.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, EKS, VPC, IAM, Lambda, S3, IoT Core, Greengrass, Bedrock, SageMaker, Kinesis, Glue, Redshift, CloudWatch), Azure (ADF, ADLS Gen2, Synapse, OpenAI, AI Search, Document Intelligence, Entra ID, Purview, Key Vault, Monitor), GCP (Vertex AI, Cloud Run, BigQuery, Cloud Storage, Document AI, Gemini, Eventarc, Cloud Functions 2nd gen)

AI and GenAI: RAG Pipelines, Agentic AI, MCP, LLM Orchestration, Multi-Agent Systems, ReAct Pattern, HITL Governance, Vector Databases (Azure AI Search, OpenSearch, Vertex AI Vector Search, Pinecone, FAISS), Embeddings, Hybrid Search, Prompt Engineering, ISO 42001 AIMS, LangGraph, LangChain, Tool Calling, Structured Outputs, Human-in-the-Loop Approval

Platform Engineering: Kubernetes (EKS, Nutanix HCI, Multi-tenant), Docker, Helm, Terraform, Infrastructure as Code, CI/CD, Jenkins, GitHub Actions, Azure DevOps, SonarQube, Trivy, Nexus, Gitea, Tuleap

Data Architecture: Medallion Lakehouse, Azure Data Factory, Synapse, Microsoft Fabric, AWS Glue, Redshift, Athena, Airflow, Apache NiFi, Kafka, Kinesis, MQTT, CDC, Schema Evolution, Data Lineage, Purview

Security and DevSecOps: Entra ID, Keycloak SSO, IAM, RBAC, KMS, Key Vault, VPC Isolation, Private Endpoints, PII Masking, Security Trimming, VAPT, Air-gapped DevSecOps, ISO 42001

Observability: Prometheus, Grafana, CloudWatch, Azure Monitor, Log Analytics, ELK/OpenSearch, Application Insights, SLO/SLA Design, BCP Frameworks

Edge AI and IoT: TensorFlow Lite, ARM Cortex, AWS IoT Core, IoT Greengrass, MQTT, EfficientDet-Lite, Embedded Systems

Programming and BI: Python (AI Pipelines, Automation, Scripting), React.js (UI Prototyping), Power BI

AWARDS AND SPEAKING

- AI Summit 2026 India — Speaker and Demonstrator: Hybrid deterministic Edge AI architectures integrating ARM platforms, TensorFlow Lite, and AWS IoT for real-time distributed intelligence.
- Embedded World 2025, Nuremberg Germany — Speaker and Demonstrator: AWS IoT Greengrass and Lambda-based distributed edge intelligence on ARM hardware at one of the world's largest embedded systems exhibitions.
- Cloud Architecture Leadership: Recognised for delivering scalable AWS edge-cloud infrastructure at Visionext 2024.
- Kubernetes Infrastructure Excellence: Recognised for architecting the ICAR national LMS Kubernetes platform supporting 1M+ learners across 76 universities.
- ISO 42001 AIMS Certification Leadership: AI policy, data quality assessments, VAPT, risk registers, HITL governance frameworks, and BCP artefacts for enterprise AI systems.

EDUCATION

MBA — Business Analytics *2022*
IMT-CDL Ghaziabad, India

B.Tech — Information Technology *2007*
IEC CET, Greater Noida (UPTU)